

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story Points)

Parameter	Details
Date	13 February 2026
Team ID	LTVIP2026TMIDS79181
Project Name	Smart Sorting – Rotten Fruit Detection using Machine Learning
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation

Sprint	Functional Requirement (Epic)	User Story No.	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Dataset Collection	USN-1	As a data analyst, I can collect and inspect fruit image datasets to understand fresh vs rotten distribution.	3	High	Data Team
Sprint-1	Image Preprocessing	USN-2	As a developer, I can preprocess images by resizing, normalization, and augmentation for model training.	4	High	ML Team
Sprint-1	Dataset Visualization	USN-3	As a team member, I can visualize dataset samples and class distribution of fresh and rotten fruits.	2	Medium	Data Team
Sprint-2	Feature Preparation	USN-4	As a developer, I can prepare image features suitable for training the machine learning model.	3	High	ML Team
Sprint-2	Model Development	USN-5	As a user, I can get fruit classification	5	High	ML Team

			results using a trained ML model.			
Sprint-2	Model Evaluation	USN-6	As a reviewer, I can view model accuracy, precision, recall, and loss metrics.	3	High	ML Team
Sprint-3	Backend API	USN-7	As a developer, I can create Flask API endpoints to upload fruit images and return prediction results.	4	High	Backend Team
Sprint-3	Model Integration	USN-8	As a developer, I can integrate the trained model with the Flask application for real-time prediction.	3	High	Backend Team
Sprint-4	Web Interface	USN-9	As a user, I can upload fruit images through a web interface and view classification results.	3	Medium	Frontend Team

Project Tracker, Velocity & Burndown Chart

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date (Actual)
Sprint-1	9	7 Days	15 Feb 2026	21 Feb 2026	9	21 Feb 2026
Sprint-2	11	7 Days	22 Feb 2026	28 Feb 2026	11	28 Feb 2026
Sprint-3	7	7 Days	01 Mar 2026	07 Mar 2026	7	07 Mar 2026
Sprint-4	5	5 Days	08 Mar 2026	12 Mar 2026	5	12 Mar 2026

Velocity

If sprint duration is **7 days** and the team completes approximately **9–11 story points per sprint**, the average velocity is approximately **1.3–1.5 story points per day**. This helps estimate development timelines for additional features such as multi-fruit detection and real-time camera integration.

Burndown Chart (Description)

A burndown chart tracks remaining story points versus time across each sprint. It helps visualize development progress, monitor delays, and ensure timely completion of the Smart Sorting system. The chart can be generated using Agile project management tools such as Jira, Trello, or Azure DevOps.

References

Atlassian Agile Project Management – Documentation on Sprints, Epics, Story Points, Velocity, and Burndown Charts.